

Computer Homework 4

Market Model, Abnormal Return and Event Study

Excerpt CNBC

<https://www.cnbc.com/2025/12/05/netflix-warner-bros-discovery-deal.html>

Netflix to buy Warner Bros. film and streaming assets in \$72 billion deal

PUBLISHED FRI, DEC 5 2025 7:11 AM EST

UPDATED FRI, DEC 5 2025 12:42 PM EST

KEY POINTS

Netflix announced Friday it's reached a deal to buy pieces of Warner Bros. Discovery, bringing a swift end to a dramatic bidding process that saw Paramount Skydance and Comcast also vying for the legacy assets.

The deal is for WBD's film studio and streaming service, HBO Max. Warner Bros. Discovery will still spin out its TV networks, which includes TNT and CNN, as previously planned.

The acquisition is expected to close in 12 to 18 months.

Excerpt CNBC

<https://www.cnbc.com/2026/02/26/warner-bros-discovery-paramount-skydance-deal-superior-netflix.html>

Netflix ditches deal for Warner Bros. Discovery after Paramount's offer is deemed superior

PUBLISHED THU, FEB 26 2026 4:20 PM EST

UPDATED FRI, FEB 27 2026 9:22 AM EST

KEY POINTS

Netflix declined to raise its bid for Warner Bros. Discovery's assets to match a revised bid by Paramount Skydance.

On Thursday, the WBD board said it valued Paramount's \$31-per-share offer to be superior to an existing deal with Netflix.

Paramount agreed to pay the \$2.8 billion breakup fee that WBD would owe Netflix if that deal didn't go through.

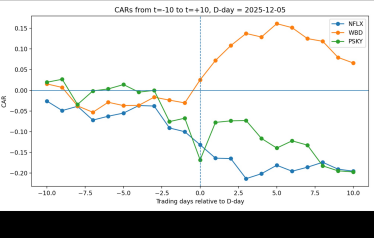
Netflix stock spiked in extended trading, while shares of Warner Bros. Discovery fell.

Please use the approach in our class to analyze the cumulative abnormal returns (CARs) of Netflix (NFLX), Warner Bros Discovery (WBD), and Paramount Skydance (PSKY). The attached Excel file contains the historical data of Netflix, Warner Bros Discovery, Paramount Skydance, and the S&P500 index levels.

1. (30%) Please use the market model ($r_{it} = \alpha_i + \beta_i r_{mt} + \varepsilon_{it}$ where r_{it} is the firm's daily return and r_{mt} is the daily market return) to estimate the alpha and beta coefficients of Netflix, Warner Bros Discovery, and Paramount Skydance before the announcement. The estimation period (estimating market model parameters) begins (252+21) trading days before the initial announcement (2024/11/4) and ends 21 trading days (2025/11/5) before the announcement. Report your regression results.
2. (15%) Please use 2025/12/5 as the D-day (t=0; the first trading day of the announcement). Compute the abnormal return on each day. Report the abnormal returns from (-10) to +10 days relative to the announcement date.
3. (20%) Please compute the cumulative abnormal returns (CARs) from -10 to +10 days relative to the announcement date (i.e., 10 trading days before the announcement to 10 days after the announcement) and plot them on a time chart. (Using the alpha and beta coefficients in (1))
4. (5%) Please compute the cumulative abnormal returns (CARs) from -1 to +1 days relative to the announcement date.
5. (20%) Next, redo (2) to (4) using 2026/2/27 as the D-day (t=0; the first trading day after the announcement date due to the time of the announcement). (Also using the alpha and beta coefficients in (1))
6. (10%) Briefly explain your findings.

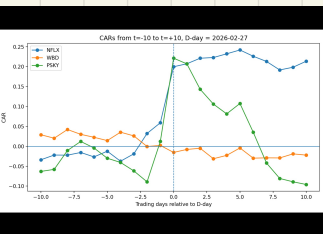
Q1_market_model_regression_results_alpha_beta									
stock	alpha	alpha_t	alpha_p_value	beta	beta_t	beta_p_value	r_squared	adj_r_squared	n_obs
NFLX	0.0201140891423000	0.000011648511900	0.984484034068000	0.871986248172000	0.786038513011510	2.2846227170372E-19	0.277703024448200	0.274217708868200	262
WBD	0.02042000911181000	0.00001624572000	0.119602026053800	0.786122100867100	0.746221717030000	2.14122044444E-19	0.277018028148100	0.273704022570100	262
PSKY	0.0148506815910000	0.001104827897200	0.5003168017018000	0.4702570686796000	0.26909160200091	0.01936010725191000	0.01465218800315700	0.01018166464017000	262

Q2_CAR_CAR_event_window_Day_2025-12-05									
Day	t	NET	NFLX event	NFLX assumed event	WBD event	WBD assumed event	PSKY event	PSKY assumed event	PSKY assumed event
2025-12-10	-10	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	-9	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	-8	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	-7	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	-6	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	-5	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	-4	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	-3	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	-2	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	-1	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	0	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	1	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	2	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	3	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	4	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	5	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	6	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	7	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	8	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	9	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2025-12-10	10	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000



Q4_CAR_minus1_to_plus1_D

stock	CAR_-1_to_+1
NFLX	-0.07301798636639050
WBD	0.09599948380258490
PSKY	-0.0019459058697200600



Q5_part4_CAR_event_window_Day_2026-03-27									
Day	t	NET	NFLX event	NFLX assumed event	WBD event	WBD assumed event	PSKY event	PSKY assumed event	PSKY assumed event
2026-03-10	-10	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	-9	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	-8	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	-7	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	-6	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	-5	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	-4	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	-3	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	-2	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	-1	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	0	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	1	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	2	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	3	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	4	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	5	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	6	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	7	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	8	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	9	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000
2026-03-10	10	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000	0.0000000000000000

Q5_part4_CAR_minus1_to_p

stock	CAR_-1_to_+1
NFLX	0.1749132755933750
WBD	-0.007360706438796250
PSKY	0.2961746741576560

The results show that the two announcements affected the three firms differently.

For the first event on 2025/12/5, WBD had a positive CAR, while NFLX had a negative CAR. This suggests that investors viewed the deal as good news for WBD, the target firm, but were concerned about the cost and risk for Netflix. PSKY's reaction was small because it was not the winning bidder at that time.

For the second event on 2026/2/27, PSKY and NFLX both had positive CARs. This suggests that investors reacted positively to Paramount's superior offer and were also relieved that Netflix did not raise its bid. Overall, the market reaction changed as new information about the bidding process became available.